

Bluestock Fintech — Mutual Fund Market Analysis

**Final Capstone Report |
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Executive Summary

This project focuses on analyzing mutual fund industry data to understand fund performance, investor behavior, market trends, and risk metrics. Data was collected from multiple mutual fund datasets and processed using Python, Pandas, and analytical techniques.

The report we are working on presents a comprehensive analysis of the Indian mutual fund industry using 10 datasets covering 40 schemes, 32,778 investor transactions, and NAV history from 2022–2026.

Key findings include:

- SBI Mutual Fund leads with ₹12.5L Cr AUM
- Monthly SIP inflows grew 2.25x from ₹12,453 Cr (2022) to ₹27,978 Cr (2025)
- Total folios doubled from 13.26 Cr to 26.12 Cr in 3 years
- Small Cap funds offer highest returns (23%) but highest risk (VaR -2.69%)
- 97% of SIP investors flagged as at-risk due to irregular intervals

The analysis revealed strong growth in Assets Under Management (AUM), increasing SIP participation, rising folio counts, and varying risk-adjusted performance across funds. Advanced analytics such as VaR, CVaR, Sharpe Ratio, and portfolio concentration analysis were also performed.

Introduction

Mutual funds have become one of the most popular investment vehicles in India. With increasing retail participation, investors require tools that can help evaluate fund performance, risk, and market trends.

The objective of this project is to build a Mutual Fund Analytics Platform that enables analysis of industry trends, fund performance, investor behavior, and portfolio risk using data-driven techniques.

Objective

1. Collect and process mutual fund datasets.
2. Perform exploratory data analysis.
3. Evaluate fund performance using financial metrics.
4. Analyze risk using advanced analytics.
5. Build an interactive Power BI dashboard.
6. Generate actionable investment insights.

Data Sources

Dataset	Records	Source
01_FUND_MASTER	40	AMFI India
02_nav_history	46000	MFAPI/AMFI
03_aum_by_fund_hous	Monthly	AMFI Factsheet
04_monthly_sip_inflows	Monthly	AMFI Factsheet
05_category_inflows	Monthly	AMFI Factsheet
06_industry_folio_cnt	Monthly	AMFI Factsheet
07_scheme_performance	40	Morningstar/AMFI
08_investor_txns	32,778	Simulated
09_portfolio_holdings	322	NSE/BSE
10_benchmark_indices	Daily	NSE

ETL Design:

Extract:

Data collected from CSV datasets and MFAPI.

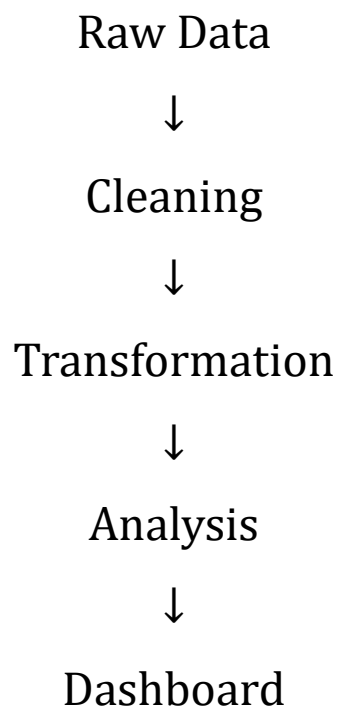
Transform:

Data cleaning, missing value handling, data type conversion, duplicate removal.

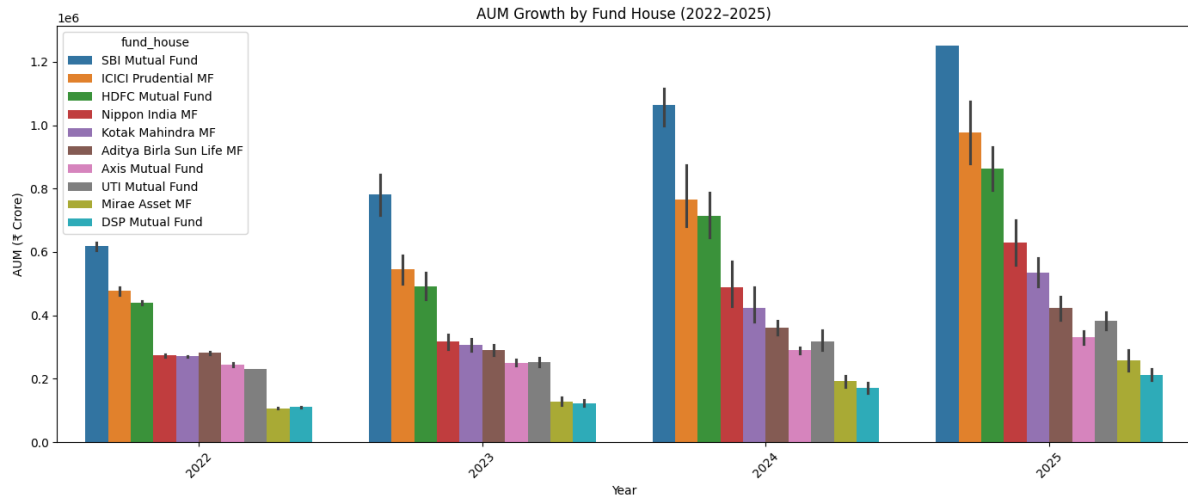
Load:

Processed datasets stored for analytics and dashboard development.

ETL flow diagram:

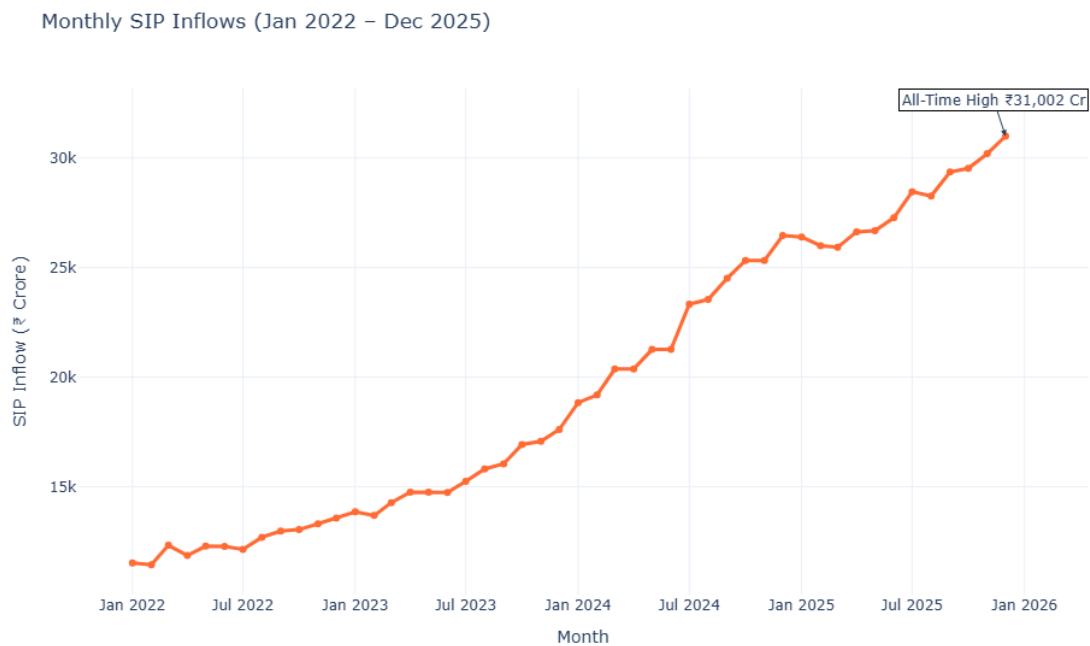


Industry AUM Analysis



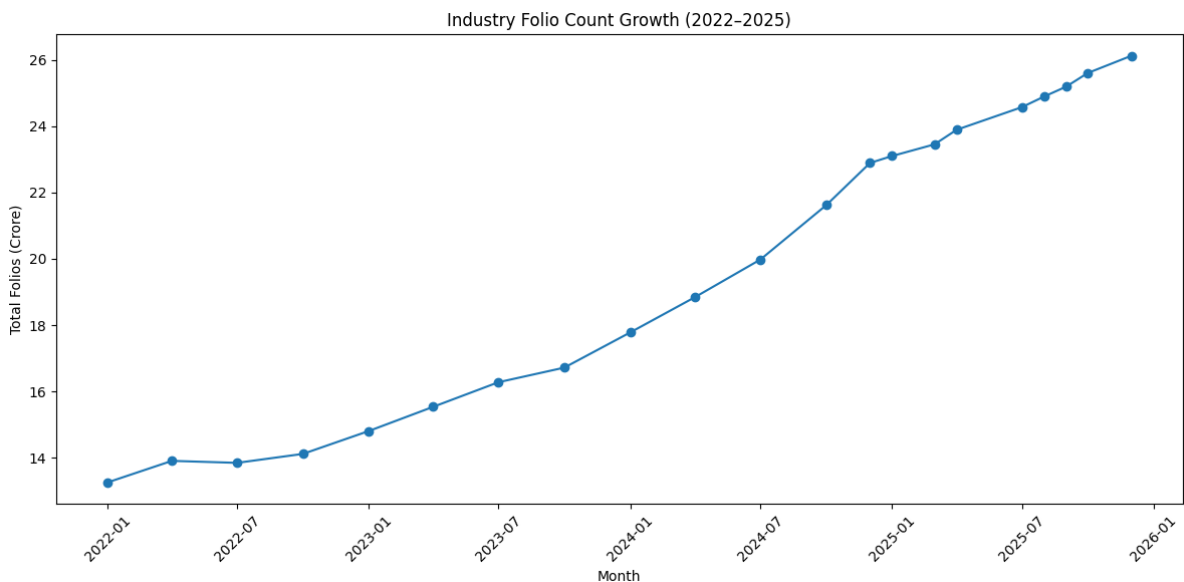
Assets Under Management (AUM) is one of the most important indicators of growth in the mutual fund industry. The analysis of industry AUM from 2022 to 2025 showed a consistent upward trend, indicating increasing investor participation and growing confidence in mutual fund investments. The growth in AUM reflects both market appreciation and continuous inflow of investor capital into mutual fund schemes.

SIP Inflow Analysis



Systematic Investment Plans (SIPs) have become the preferred investment mode for retail investors. Analysis of monthly SIP inflows revealed a steady increase throughout the study period, with inflows reaching record highs by the end of 2025. Rising SIP contributions indicate increasing awareness of disciplined investing and long-term wealth creation among investors.

Folio Growth Analysis



Folio count represents the number of investor accounts maintained across mutual fund schemes. The analysis showed substantial growth in folio counts between 2022 and 2025, reflecting expansion of the investor base. This trend demonstrates increasing adoption of mutual funds across different investor segments and regions.

Asset Management Company Analysis

The comparison of Assets Under Management across Asset Management Companies (AMCs) highlighted differences in market leadership.

Large AMCs such as SBI Mutual Fund, HDFC Mutual Fund, and ICICI Prudential Mutual Fund managed a significant share of industry assets.

The analysis helps understand market concentration and competitive positioning within the mutual fund industry.

Fund Performance Analytics

CAGR table -

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Fund Scorecard Table -

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	amfi_code	cagr	cagr_pct	sharpe_ra	alpha	beta	max_draw	return_rar	sharpe_ra	alpha_ran	drawdown	fund_score			
2	120505	0.328015	32.80159	1.180101	0.102647	1.194814	-0.18189	1	5	5	16	85.5902777777777			
3	148567	0.309499	30.94992	1.448291	0.181967	0.557134	-0.11266	5	1	1	33	81.7708333333333			
4	148569	0.319244	31.92448	1.234930	0.09647	1.174586	-0.16397	4	3	8	20	80.7291666666667			
5	120843	0.308833	30.88332	1.306743	0.102644	1.068745	-0.12974	6	2	6	28	78.125			
6	100033	0.300997	30.09970	1.093698	0.114423	0.991589	-0.16217	7	7	4	21	76.5625			
7	119551	0.257849	25.78492	1.208266	0.141513	0.562930	-0.15012	10	4	2	24	76.3888888888889			
8	149323	0.295581	29.55810	1.132121	0.08772	1.120482	-0.17248	8	6	9	19	73.6111111111111			
9	120504	0.232774	23.27744	1.026524	0.115994	0.606529	-0.12588	13	10	3	29	65.625			
10	118632	0.240311	24.03119	1.081659	0.057106	1.011984	-0.17414	11	8	16	18	64.2361111111111			
11	149324	0.322621	32.26210	0.949795	-0.05128	2.214803	-0.31172	3	13	29	4	64.0625			
12	101206	0.235204	23.52048	1.027213	0.09744	0.737015	-0.11292	12	9	7	32	63.7152777777778			
13	119598	0.323980	32.39808	0.945308	-0.08813	2.457368	-0.28706	2	14	31	5	62.5			
14	148568	0.224998	22.49984	0.926287	0.07904	0.810411	-0.15273	14	15	10	22	57.8125000000001			
15	119094	0.281926	28.19260	0.998230	0.01687	1.520896	-0.20961	9	11	27	13	57.8125			
16	119552	0.215080	21.50802	0.953279	0.06197	0.85463	-0.11803	15	12	14	30	53.8194444444444			
17	102885	0.182188	18.21882	0.817099	0.07188	0.616292	-0.1086	18	16	11	34	47.9166666666667			
18	125497	0.190036	19.00361	0.775790	0.03372	0.912326	-0.152	16	18	21	23	45.1388888888889			
19	120503	0.187713	18.77133	0.797972	0.02873	0.924792	-0.13898	17	17	23	27	42.1875000000001			
20	118633	0.162622	16.26221	0.645207	0.04423	0.698535	-0.1863	22	21	18	15	41.1458333333333			
21	120506	0.170722	17.07222	0.648878	0.018469	0.914058	-0.19826	19	20	26	14	39.9305555555556			
22	102887	0.168286	16.82867	0.619517	0.024546	0.868286	-0.2154	20	22	24	11	39.5833333333333			
23	118635	0.159454	15.94548	0.66486	0.05179	0.625842	-0.11651	23	19	17	31	36.9791666666667			
24	120841	0.135223	13.52234	0.494272	0.020996	0.69518	-0.17574	24	24	25	17	30.9027777777778			
25	120507	0.07232	7.232410	0.495723	0.06836	-0.00573	-0.00098	29	23	12	40	27.6041666666667			
26	118634	0.163613	16.36139	0.448434	-0.17835	2.24202	-0.23345	21	26	37	9	26.7361			
27	149322	0.133835	13.38352	0.464037	-0.01926	0.948724	-0.14845	25	25	28	25	24.1319444444444			
28	125498	0.112936	11.29366	0.300611	-0.0985	1.375060	-0.21117	26	27	34	12	21.7013888888889			
29	119093	0.07892	7.891944	0.129614	-0.07812	1.013926	-0.21751	28	29	30	10	21.3541666666667			
30	101207	0.07933	7.933120	0.162660	-0.35074	2.878184	-0.35447	27	28	40	3	18.75			
31	120844	0.0691	6.91038	-0.08875	0.06624	-0.01067	-0.00116	30	34	13	39	16.6666666666667			
32	120842	0.06697	6.696806	0.076	-0.09246	1.075704	-0.24004	31	30	33	8	15.9722222222222			
33	101208	0.06504	6.504429	-0.81557	0.06103	-0.00102	-0.00162	32	40	15	38	8.33333333333333			
34	119120	0.05885	5.885271	-0.22658	0.04251	0.08487	-0.04329	34	37	19	36	6.77083333333333			
35	100016	0.026352	2.635245	-0.20152	-0.0897	0.78844	-0.24734	37	35	32	7	6.42361111111111			
36	119599	0.02053	2.053064	-0.05719	-0.27362	2.039654	-0.52574	38	32	38	1	5.90277777777778			
37	119092	0.06363	6.363494	0.03078	-0.11502	1.159021	-0.14402	33	31	35	26	5.38194444444444			
38	118636	0.053131	5.313155	-0.35663	0.034952	0.09957	-0.08316	35	38	20	35	4.51389			
39	119095	0.015204	1.520455	-0.07597	-0.30562	2.210849	-0.51678	39	33	39	2	2.95139			
40	100025	0.04455	4.455091	-0.56709	0.0332	0.060713	-0.04308	36	39	22	37	0.520833333333334			
41	102886	0.01171	1.17094	-0.20583	-0.11568	0.901441	-0.28001	40	36	36	6	0			
42															
43															

Fund performance was evaluated using multiple return and risk-adjusted metrics. Compound Annual Growth Rate (CAGR) was calculated for different investment horizons to measure long-term growth. The analysis helped identify funds that consistently delivered superior returns over time while accounting for market fluctuations.

Risk-Adjusted Performance Analysis

Sharpe Ratio ranking -

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	amfi_coc	sharpe_ratio											
2	148567	1.4483											
3	120843	1.3067											
4	148569	1.2349											
5	119551	1.2083											
6	120505	1.1801											
7	149323	1.1321											
8	100033	1.0937											
9	118632	1.0817											
10	101206	1.0272											
11	120504	1.0265											
12	119094	0.9982											
13	119552	0.9533											
14	149324	0.9498											
15	119598	0.9453											
16	148568	0.9263											
17	102885	0.8171											
18	120503	0.798											
19	125497	0.7758											
20	118635	0.6649											
21	120506	0.6489											
22	118633	0.6452											
23	102887	0.6195											
24	120507	0.4957											
25	120841	0.4943											
26	149322	0.464											
27	118634	0.4484											
28	125498	0.3006											
29	101207	0.1627											
30	119093	0.1296											
31	120842	0.076											
32	119092	0.0308											
33	119599	-0.0572											
34	119095	-0.076											
35	120844	-0.0887											
36	100016	-0.2015											
37	102886	-0.2058											
38	119120	-0.2266											
39	118636	-0.3566											
40	100025	-0.5671											
41	101208	-0.8156											
42													
43													
44													
45													
46													

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sharpe_ratio

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Sortino Ratio ranking -

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	amfi co	sortino_ratio											
2	148567	2.3856											
3	120843	2.3643											
4	148569	2.1469											
5	119551	2.1403											
6	120505	2.0294											
7	149323	1.8751											
8	118632	1.8501											
9	100033	1.8291											
10	120504	1.8053											
11	101206	1.7996											
12	119094	1.7038											
13	119598	1.6753											
14	148568	1.6327											
15	149324	1.6198											
16	119552	1.6098											
17	102885	1.4357											
18	120503	1.3379											
19	125497	1.3307											
20	118633	1.1545											
21	118635	1.138											
22	120506	1.0959											
23	102887	1.0888											
24	120507	1.0521											
25	120841	0.8626											
26	149322	0.7621											
27	118634	0.746											
28	125498	0.5088											
29	101207	0.2766											
30	119093	0.2211											
31	120842	0.1247											
32	119092	0.0526											
33	119599	-0.0944											
34	119095	-0.1282											
35	120844	-0.1877											
36	102886	-0.3472											
37	100016	-0.351											
38	119120	-0.3768											
39	118636	-0.6135											
40	100025	-0.9418											
41	101208	-1.681											
42													
43													
44													
45													
46													

Sharpe Ratio and Sortino Ratio were used to evaluate risk-adjusted returns. While Sharpe Ratio considers total volatility, Sortino Ratio focuses only on downside risk. Funds with higher values demonstrated better return generation relative to the risk undertaken by investors.

Market Risk Analysis

alpha_beta.csv -

	A	B	C	D	E	F	G	H	I	J	K
1	amfi_code	alpha	beta								
2	100016	-0.0897	0.78844								
3	100025	0.0332	0.06071								
4	100033	0.11442	0.99159								
5	101206	0.09744	0.73702								
6	101207	-0.35074	2.87818								
7	101208	0.06103	-0.00102								
8	102885	0.07188	0.61629								
9	102886	-0.11568	0.90144								
10	102887	0.02455	0.86829								
11	118632	0.05711	1.01198								
12	118633	0.04423	0.69854								
13	118634	-0.17835	2.24202								
14	118635	0.05179	0.62584								
15	118636	0.03495	0.09957								
16	119092	-0.11502	1.15902								
17	119093	-0.07812	1.01393								
18	119094	0.01687	1.5209								
19	119095	-0.30562	2.21085								
20	119120	0.04251	0.08487								
21	119551	0.14151	0.56293								
22	119552	0.06197	0.85463								
23	119598	-0.08813	2.45737								
24	119599	-0.27362	2.03965								
25	120503	0.02873	0.92479								
26	120504	0.11599	0.60653								
27	120505	0.10265	1.19481								
28	120506	0.01847	0.91406								
29	120507	0.06836	-0.00573								
30	120841	0.021	0.69518								
31	120842	-0.09246	1.0757								
32	120843	0.10264	1.06875								
33	120844	0.06624	-0.01067								
34	125497	0.03372	0.91233								
35	125498	-0.0985	1.37506								
36	148567	0.18197	0.55713								
37	148568	0.07904	0.81041								
38	148569	0.09647	1.17459								
39	149322	-0.01926	0.94872								
40	149323	0.08772	1.12048								
41	149324	-0.05128	2.2148								
42											

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alpha_beta

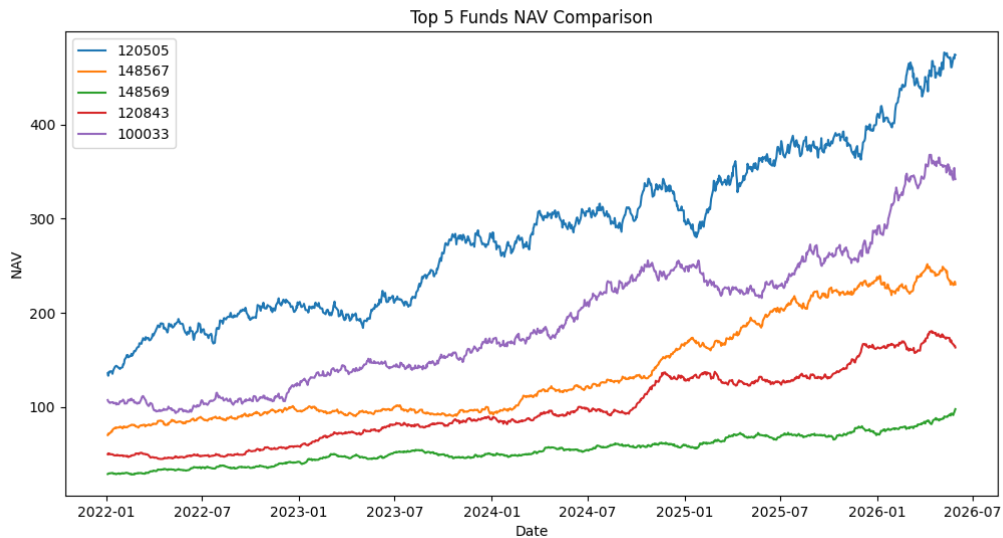
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max drawdown results -

	A	B	C	D	E	F	G	H	I	J	K	L
1	amfi_coc	max_drawdown										
2	119599	-0.5257										
3	119095	-0.5168										
4	101207	-0.3545										
5	149324	-0.3117										
6	119598	-0.2871										
7	102886	-0.28										
8	100016	-0.2473										
9	120842	-0.24										
10	118634	-0.2334										
11	119093	-0.2175										
12	102887	-0.2154										
13	125498	-0.2112										
14	119094	-0.2096										
15	120506	-0.1983										
16	118633	-0.1863										
17	120505	-0.1819										
18	120841	-0.1757										
19	118632	-0.1741										
20	149323	-0.1725										
21	148569	-0.164										
22	100033	-0.1622										
23	148568	-0.1527										
24	125497	-0.152										
25	119551	-0.1501										
26	149322	-0.1484										
27	119092	-0.144										
28	120503	-0.139										
29	120843	-0.1297										
30	120504	-0.1259										
31	119552	-0.118										
32	118635	-0.1165										
33	101206	-0.1129										
34	148567	-0.1127										
35	102885	-0.1086										
36	118636	-0.0832										
37	119120	-0.0433										
38	100025	-0.0431										
39	101208	-0.0016										
40	120844	-0.0012										
41	120507	-0.001										
42												
43												

Alpha and Beta were calculated to understand fund behavior relative to benchmark indices. Alpha measured excess returns generated beyond benchmark performance, while Beta measured sensitivity to market movements. Maximum Drawdown was used to estimate the worst historical decline experienced by a fund during the study period.

Benchmark Comparison



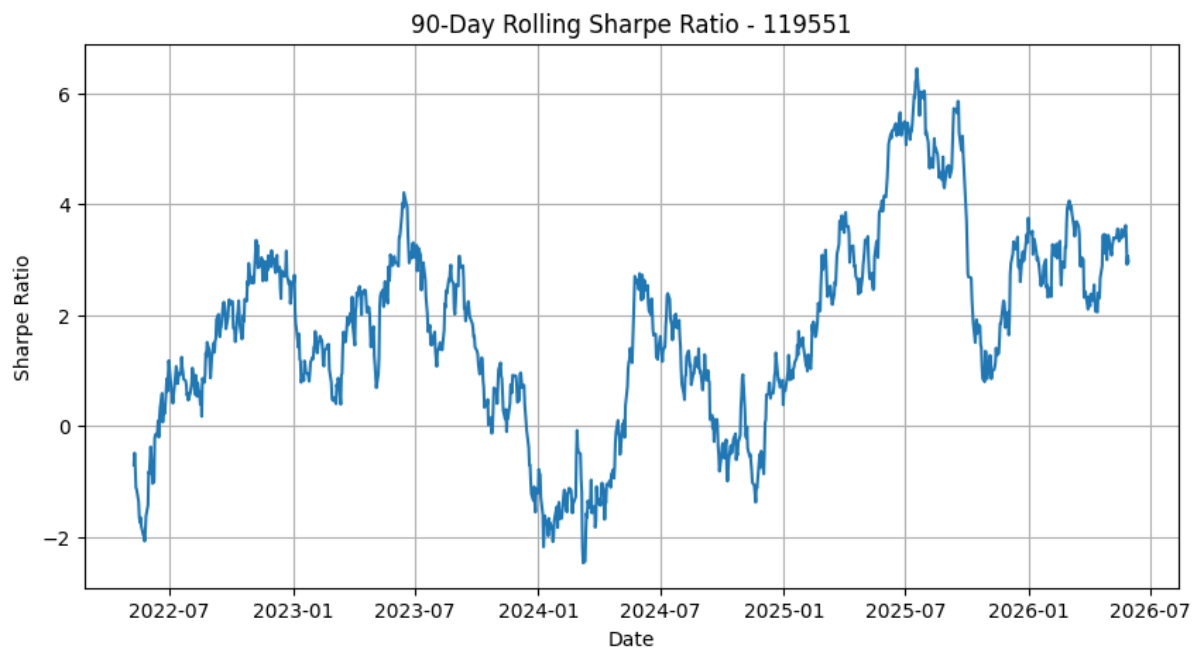
Top-performing funds were compared against benchmark indices such as Nifty 50 and Nifty 100. The comparison enabled evaluation of whether active fund management generated returns beyond market performance. Tracking error analysis was also performed to measure deviations from benchmark returns.

Advanced Analytics

VaR - CVaR results -

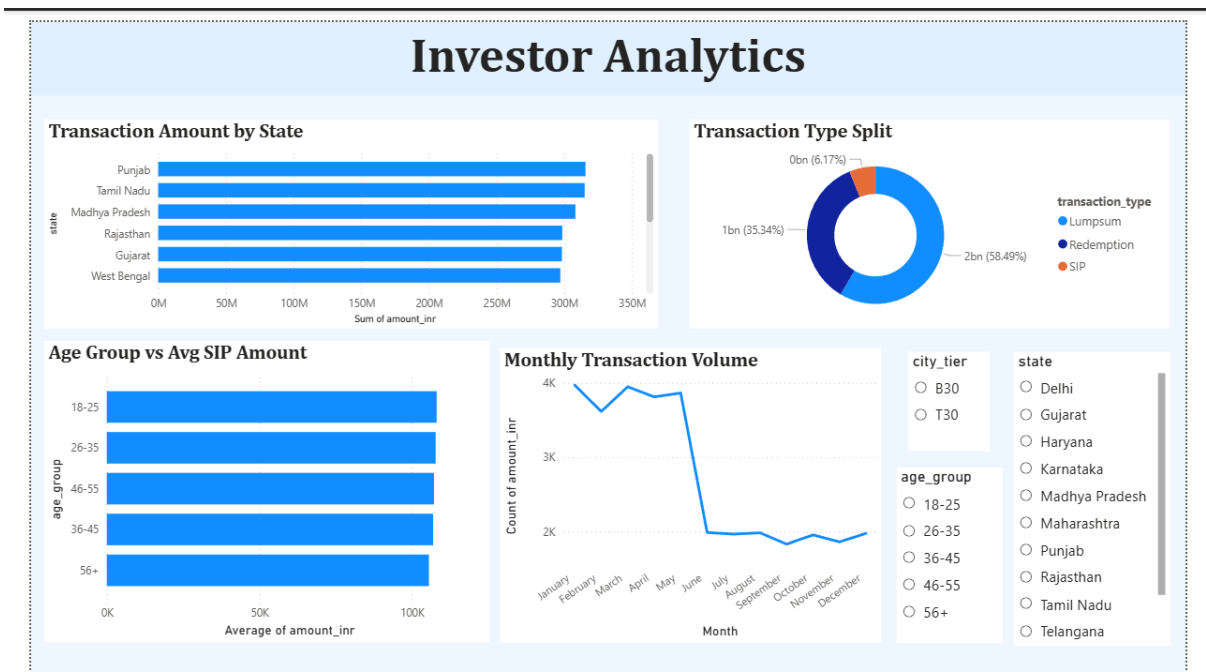
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Rolling Sharpe -



Advanced risk metrics including Value at Risk (VaR) and Conditional Value at Risk (CVaR) were computed to estimate downside risk. Rolling 90-day Sharpe Ratio analysis was performed to evaluate consistency of risk-adjusted performance over time. These metrics provided a deeper understanding of portfolio risk characteristics.

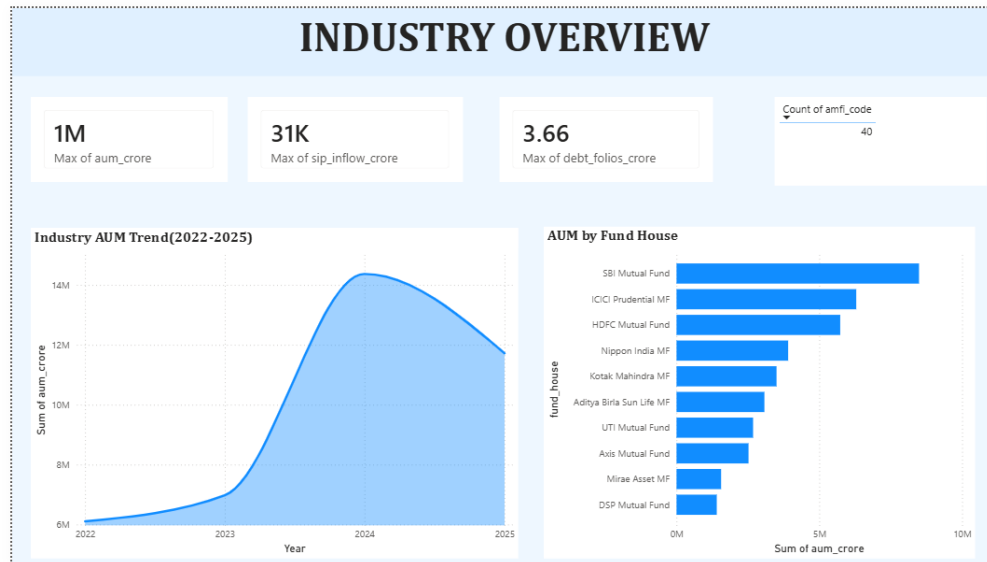
Investor Analytics



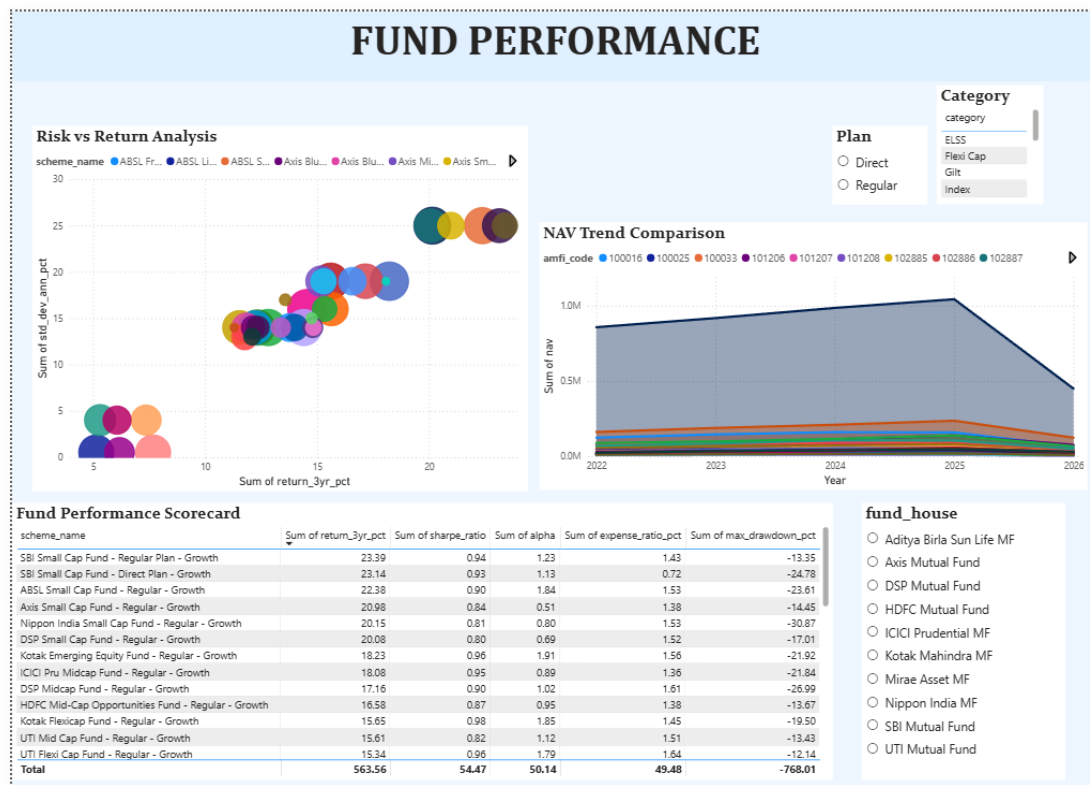
Investor transaction data was analyzed to understand demographic participation, geographic distribution, and investment behavior. The analysis highlighted variations in investment patterns across age groups, states, and investor categories. These insights can help fund houses design targeted investment products and marketing strategies.

Dashboard Development

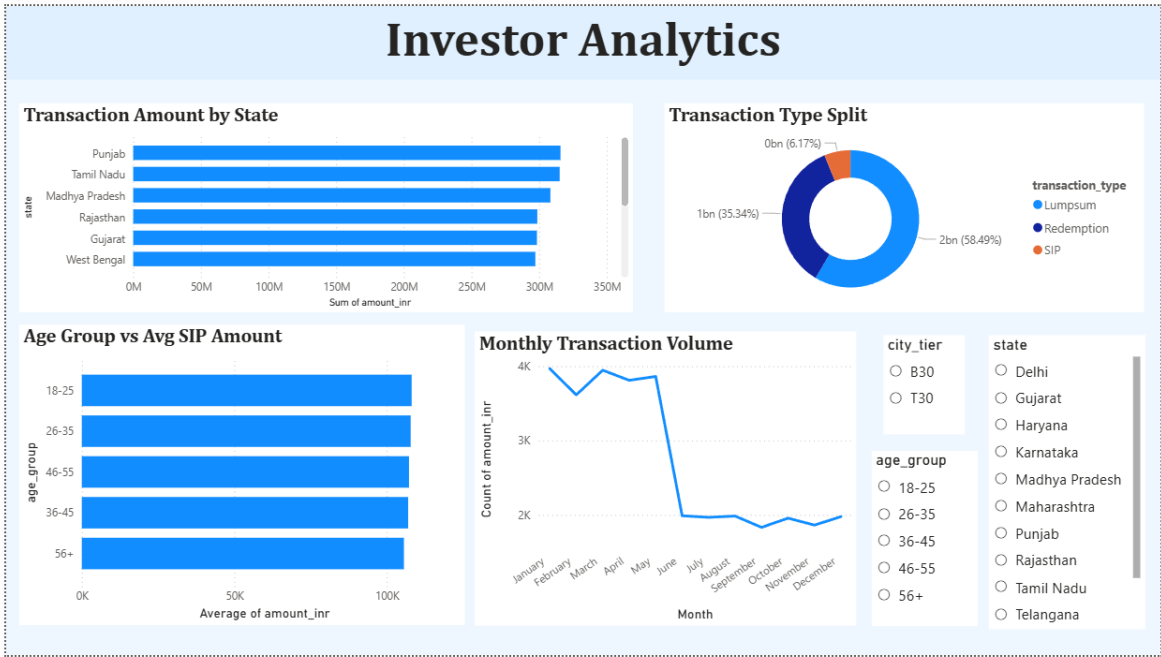
INDUSTRY OVERVIEW -



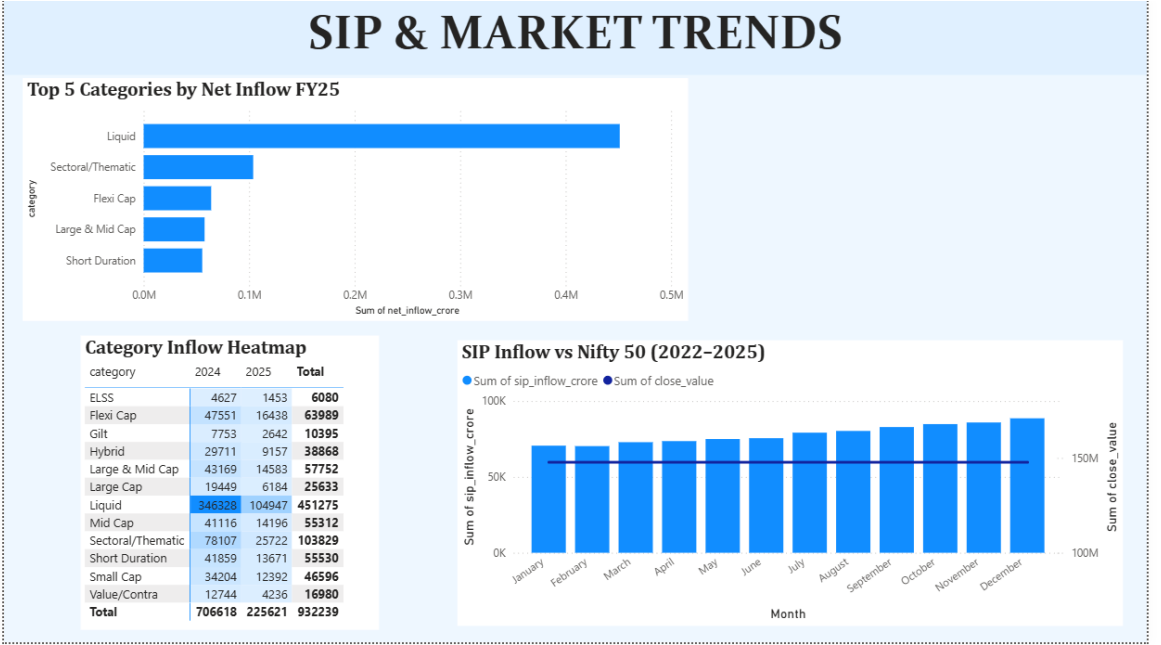
FUND PERFORMANCE -



INVESTOR ANALYTICS -



SIP&MARKET TRENDS -



An interactive Power BI dashboard was developed to provide a comprehensive view of mutual fund industry trends, fund performance, investor behavior, and SIP growth. The dashboard consists of multiple pages with filters, KPI cards, charts, and drill-through functionality. Users can explore data dynamically and gain actionable insights through interactive visualizations.

Key Findings:

- Industry Assets Under Management (AUM) showed strong growth between 2022 and 2025.
- SIP inflows increased consistently, indicating rising investor confidence and participation.
- Equity-oriented funds attracted the highest investor interest and folio growth.
- Funds with higher Sharpe Ratios generally delivered superior risk-adjusted returns.
- Portfolio concentration varied significantly across schemes, highlighting diversification differences.
- Advanced risk metrics revealed variations in downside risk among funds.
- Power BI dashboards enabled efficient exploration of industry and investor trends.

Recommendations and Limitations

Recommendations:

- Investors should evaluate both returns and risk before selecting funds.
- Diversification across sectors and categories should be maintained.
- SIP investing should be encouraged for long-term wealth creation.
- Portfolio concentration should be monitored regularly to reduce risk exposure.

Limitations:

- Analysis was based on available historical data.
- Market conditions may change over time and affect future performance.
- Recommendations are based on historical trends and should not be treated as investment advice.

Conclusion

The Mutual Fund Analytics Platform successfully demonstrated the complete data analytics lifecycle, including data collection, cleaning, exploratory analysis, performance evaluation, risk assessment, dashboard development, and insight generation. The project provided valuable understanding of mutual fund performance, investor behavior, and industry trends while showcasing practical application of Python, Power BI, and financial analytics techniques. The resulting dashboards and analytical outputs can support informed decision-making for investors and financial institutions.